Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Module 2
Linear Transformation and Matrix Decomposition

Linear Independence

Understanding Linear Independence in Vector Spaces
Reading • 18 min

Linear Independence, Basis, and Rank
Ungraded App Item • 20 min

Linear Independence
Reading • 30 min

Linear Mapping

Exploring Transformations: Understanding Linear Mappings
Reading • 1h

Linear Mapping: Part 1
Video • 8 min

Linear Mapping: Part 2
Video • 8 min

Linear Mapping
Reading • 5 min

Matrix Representation
Reading • 6 min

Practice Quiz: Linear Mapping
Graded Assignment • 18 min

Lesson 3: Eigenvalues and Eigenvectors

Introduction to Eigenvalues and Eigenvectors
Reading • 40 min

Eigenvectors and Eigenvalues: Examples and Applications
Reading • 30 min

Lesson 4: Cholesky Decomposition

Cholesky Decomposition
Reading • 5 min

Solved Examples
Reading • 10 min

Lesson 5: Singular Value Decomposition

Introduction to Singular Value Decomposition (SVD)
Reading • 28 min

Derivation of Singular Value Decomposition (SVD) from Eigenvalues and Eigenvectors
Reading • 15 min

Module 3
Probability Foundations for Statistical Learning

Module 4
Introduction to Model Evaluation and Optimization

Exploring Transformations: Understanding Linear Mappings

0:06 / 0:58 2x

Read Section 2.7: “Linear Mappings” from Mathematics for Machine Learning by Marc Peter Deisenroth, A. Aldo Faisal, and Cheng Soon Ong.

In this section, you will learn the concept of linear mappings, which are essential for understanding how transformations between vector spaces can be represented and manipulated using matrices. You will learn how to represent linear mappings with matrices, perform basis changes, and apply these transformations to analyze and solve problems in vector spaces.

As you read, consider the following questions:

- How can linear mappings between vector spaces be represented using matrices?
- What is the process of changing bases in the context of linear mappings, and why is it important?
- How do linear mappings help in understanding and solving complex problems in machine learning and other applications?
- What are the practical implications of linear transformations in the manipulation and analysis of data?

Go to next item ✓ Completed

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